

scVI human primary-use consensus

Independent backup of the reconciled human reference labels

Scope

This document preserves the 50 formal-validation primary-use labels. Machine labels are not included. Forty records are used in the primary-use score; ten are retained only for coverage, citation, or no-final process auditing.

Label set: Mention only; Apply to biological data; Benchmark/evaluate; Extend/develop.

Consensus records

Review ID and article	Human consensus primary use	Scoring scope and provenance
HV-001 Computational tools for analyzing single-cell data in pluripotent cell differentiation studies DOI: 10.1016/j.crmeth.2021.100087	Mention only	Process audit only Reviewer 1 record
HV-002 Multimodal Hypersampling of the adult human heart resolves tissue-specific cell states and disease-associated variants DOI: 10.64898/2026.06.16.732617	Apply to biological data	Included in primary-use score Reviewer 1 record
HV-003 BrainBridge Characterizes Key Factors affecting Alzheimer's Disease and Associated Phenotypes DOI: 10.64898/2025.12.03.692255	Benchmark/evaluate	Included in primary-use score User-directed: Reviewer 1
HV-004 Hybrid deep learning–mechanistic modeling of cellular dynamics from a spatiotemporal single-cell atlas DOI: 10.1101/2025.10.02.679951	Mention only	Included in primary-use score Reviewer 1 record
HV-005 Single-cell integration and multi-modal profiling reveals phenotypes and spatial organization of neutrophils in colorectal cancer. DOI: 10.1016/j.ccell.2025.12.003	Apply to biological data	Included in primary-use score Reviewer 1 record

Review ID and article	Human consensus primary use	Scoring scope and provenance
HV-006 Implementing distinct spatial proteogenomic technologies: opportunities, challenges, and key considerations DOI: 10.1093/cei/uxae077	Mention only	Included in primary-use score Reviewer 1 record
HV-007 Biological network-inspired interpretable variational autoencoder DOI: 10.1101/2020.12.17.423310	Mention only	Included in primary-use score Reviewer 1 record
HV-008 LAIOR: a hyperbolic neural ODE variational framework for interpretable single-cell manifold learning and trajectory inference DOI: 10.3389/fgene.2026.1838613	Benchmark/evaluate	Included in primary-use score User-directed: Benchmark
HV-009 Coding agents author interpretable single-cell embedding models from the literature DOI: 10.64898/2026.07.07.737048	Benchmark/evaluate	Included in primary-use score Reviewer 1 record
HV-010 Single-cell characterization of anti-LAG-3 and anti-PD-1 combination treatment in patients with melanoma DOI: 10.1172/jci164809	Apply to biological data	Included in primary-use score Reviewer 1 record
HV-011 JIND-Multi: Leveraging Multiple Labeled Datasets for Automated Annotation of Single-Cell RNA and ATAC Data DOI: 10.1101/2025.01.15.633130	Mention only	Included in primary-use score Reviewer 1 record
HV-012 sCellIST: a Multiple Instance Learning approach to predict single-cell gene expression from H&E images using spatial transcriptomics DOI: 10.1101/2024.11.07.622225	Extend/develop	Process audit only User-directed: Reviewer 1
HV-013 COVID-Datathon: Biomarker identification for COVID-19 severity based on BALF scRNA-seq data DOI: 10.48550/arxiv.2110.04986	Mention only	Included in primary-use score Reviewer 1 record

Review ID and article	Human consensus primary use	Scoring scope and provenance
HV-014 Kinetic Foundation of the Zero-Inflated Negative Binomial Model for Single-Cell RNA Sequencing Data DOI: 10.1137/19m1253198	Mention only	Process audit only Reviewer 1 record
HV-015 Recovering Single-cell Heterogeneity Through Information-based Dimensionality Reduction DOI: 10.1101/2021.01.19.427303	Benchmark/evaluate	Included in primary-use score Reviewer 1 record
HV-016 AtlasLens: Metadata-centric exploration and analysis of single-cell atlases DOI: 10.64898/2026.07.10.735823	Apply to biological data	Included in primary-use score User-directed: Reviewer 1
HV-017 Multi-omic identification of perineurial hyperplasia and lipid-associated nerve macrophages in human polyneuropathies DOI: 10.1038/s41467-025-62964-8	Apply to biological data	Included in primary-use score Reviewer 1 record
HV-018 Hierarchical classification of immune cell transcriptomes at population-scale DOI: 10.64898/2026.05.30.728980	Apply to biological data	Included in primary-use score Reviewer 1 record
HV-019 Anti-bias training for (sc)RNA-seq: experimental and computational approaches to improve precision DOI: 10.1093/bib/bbab148	Mention only	Included in primary-use score Reviewer 1 record
HV-020 SCIntRuler: guiding the integration of multiple single-cell RNA-seq datasets with a novel statistical metric DOI: 10.1093/bioinformatics/btae537	Mention only	Included in primary-use score Reviewer 1 record
HV-021 Biologically Informed Variational Inference Enables Interpretable Cell Phenotyping and Discovery DOI: 10.1101/2025.06.10.657924	Extend/develop	Process audit only User-directed: Reviewer 1

Review ID and article	Human consensus primary use	Scoring scope and provenance
HV-022 Transforming microfluidics for single-cell analysis with robotics and artificial intelligence DOI: 10.1039/d5lc00216h	Mention only	Included in primary-use score Reviewer 1 record
HV-023 Transcriptomic advances in studies of muscle stem cell aging: From bulk to single-cell and beyond DOI: 10.1038/s41422-026-01240-w	Mention only	Included in primary-use score Reviewer 1 record
HV-024 A multi-dataset single-cell meta-analysis of human keloid skin across Asian, Black and White populations reveals endothelial and mesenchymal programs linked to ethnic disparities DOI: 10.3389/fmed.2026.1751725	Apply to biological data	Process audit only Reviewer 1 record
HV-025 PI3Ky in B cells promotes antibody responses and generation of antibody-secreting cells DOI: 10.1038/s41590-024-01890-1	Apply to biological data	Included in primary-use score Reviewer 1 record
HV-026 Tumor-infiltrating CD27 - IgD - regulatory B cells suppress cytotoxic CD8 + T cell responses in renal cell carcinoma DOI: 10.1101/2025.07.08.663720	Apply to biological data	Included in primary-use score Reviewer 1 record
HV-027 Multi-modal Diffusion Model with Dual-Cross-Attention for Multi-Omics Data Generation and Translation DOI: 10.1101/2025.02.27.640020	Benchmark/evaluate	Included in primary-use score Reviewer 1 record
HV-028 Deep learning applications in single-cell genomics and transcriptomics data analysis DOI: 10.1016/j.biopha.2023.115077	Mention only	Included in primary-use score Reviewer 1 record

Review ID and article	Human consensus primary use	Scoring scope and provenance
HV-029 Relevant Applications of Generative Adversarial Networks in Drug Design and Discovery: Molecular De Novo Design, Dimensionality Reduction, and De Novo Peptide and Protein Design DOI: 10.3390/molecules25143250	Mention only	Included in primary-use score Reviewer 1 record
HV-030 Epigenetic biomarkers in inflammatory bowel diseases—computational challenges and opportunities DOI: 10.1093/ecco-jcc/jjag022	Mention only	Included in primary-use score Reviewer 1 record
HV-031 Candida albicans exhibits distinct cytoprotective responses to anti-fungal drugs that facilitate the evolution of drug resistance DOI: 10.1101/2020.01.21.914549	Apply to biological data	Included in primary-use score Reviewer 1 record
HV-032 Effectively Clustering Single Cell RNA Sequencing Data by Sparse Representation DOI: 10.1109/tcbb.2021.3128576	Mention only	Included in primary-use score Reviewer 1 record
HV-033 HoloCell: A Generative Foundation Model for Holistic Cellular Modeling DOI: 10.64898/2026.06.07.730684	Benchmark/evaluate	Included in primary-use score Reviewer 1 record
HV-034 Single-Cell Multi-Omics: Insights into Therapeutic Innovations to Advance Treatment in Cancer DOI: 10.3390/ijms26062447	Mention only	Included in primary-use score Reviewer 1 record
HV-035 Sequencing dropout-and-batch effect normalization for single-cell mRNA profiles: a survey and comparative analysis DOI: 10.1093/bib/bbaa248	Benchmark/evaluate	Process audit only Reviewer 1 record
HV-036 SECANT: a biology-guided semi-supervised method for clustering, classification, and annotation of single-cell multi-omics DOI: 10.1093/pnasnexus/pgac165	Extend/develop	Included in primary-use score Reviewer 1 record

Review ID and article	Human consensus primary use	Scoring scope and provenance
HV-037 Assessing Markovian and Delay Models for Single-Nucleus RNA Sequencing DOI: 10.1007/s11538-023-01213-9	Mention only	Process audit only User-directed: Reviewer 1
HV-038 Deep-learning methods for unveiling large-scale single-cell transcriptomes DOI: 10.20892/j.issn.2095-3941.2023.0436	Mention only	Included in primary-use score Reviewer 1 record
HV-039 SCOPE: Localizing fate-decision states and their regulatory drivers in single-cell differentiation DOI: 10.64898/2026.04.07.717037	Apply to biological data	Included in primary-use score User-directed: Reviewer 1
HV-040 Best practices for single-cell analysis across modalities DOI: 10.1038/s41576-023-00586-w	Mention only	Process audit only User-directed: Reviewer 1
HV-041 Decoding disease and therapy through multiomics integration and systems analysis DOI: 10.1093/bib/bbag311	Mention only	Included in primary-use score Reviewer 1 record
HV-042 Pertpy: an end-to-end framework for perturbation analysis DOI: 10.1038/s41592-025-02909-7	Extend/develop	Included in primary-use score User-directed: Reviewer 2
HV-043 Benchmarking atlas-level data integration in single-cell genomics DOI: 10.1038/s41592-021-01336-8	Benchmark/evaluate	Included in primary-use score Reviewer 1 record
HV-044 Toward universal cell embeddings: integrating single-cell RNA-seq datasets across species with SATURN DOI: 10.1038/s41592-024-02191-z	Benchmark/evaluate	Included in primary-use score Reviewer 1 record
HV-045 An overview of computational methods in single-cell transcriptomic cell type annotation DOI: 10.1093/bib/bbaf207	Mention only	Included in primary-use score Reviewer 1 record

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HV-046 Projecting single-cell transcriptomics data onto a reference T cell atlas to interpret immune responses DOI: 10.1101/2020.06.23.166546	Mention only	Process audit only User-directed: Reviewer 1
HV-047 Deep learning in single-cell and spatial transcriptomics data analysis: advances and challenges from a data science perspective DOI: 10.1093/bib/bbaf136	Benchmark/evaluate	Included in primary-use score User-directed: Reviewer 2
HV-048 Beyond the List: Bioagent-Agnostic Signatures Could Enable a More Flexible and Resilient Biodefense Posture Than an Approach Based on Priority Agent Lists Alone DOI: 10.3390/pathogens10111497	Mention only	Process audit only Reviewer 1 record
HV-049 Trends and Potential of Machine Learning and Deep Learning in Drug Study at Single-Cell Level DOI: 10.34133/research.0050	Mention only	Included in primary-use score Reviewer 1 record
HV-050 Batch correcting single-cell spatial transcriptomics count data with Crescendo improves visualization and detection of spatial gene patterns DOI: 10.1186/s13059-025-03479-9	Benchmark/evaluate	Included in primary-use score Reviewer 1 record